

# Conditional Statements

## 1 Simple Decision (If Only)

### Q1. Positive Number

**Problem:** Check whether a number is positive.

**Algorithm:**

1. Start
2. Take a number as input
3. Check if the number is greater than 0
4. If yes, display “Positive number”
5. Stop

### Q2. Excellent Performance

**Problem:** Display “Excellent” if marks are 90 or above.

**Algorithm:**

1. Start
2. Take marks as input
3. Check if marks are greater than or equal to 90
4. If yes, display “Excellent”
5. Stop

### Q3. Senior Citizen Check

**Problem:** Display “Senior Citizen” if age is 60 or above.

**Algorithm:**

1. Start

2. Take age as input
3. Check if age is greater than or equal to 60
4. If yes, display “Senior Citizen”
5. Stop

## 2 Two-Way Decision (If–Else)

### Q4. Even or Odd

**Problem:** Determine whether a number is even or odd.

**Algorithm:**

1. Start
2. Take a number as input
3. Check if the number is divisible by 2
4. If yes, display “Even number”
5. Otherwise, display “Odd number”
6. Stop

### Q5. Temperature Status

**Problem:** Decide whether the weather is hot or cold.

**Algorithm:**

1. Start
2. Take temperature as input
3. Check if temperature is greater than 30
4. If yes, display “Hot weather”
5. Otherwise, display “Cold weather”
6. Stop

## Q6. Login Status

**Problem:** Check whether a user is logged in.

**Algorithm:**

1. Start
2. Take login status as input
3. Check if user is logged in
4. If yes, display “Access granted”
5. Otherwise, display “Access denied”
6. Stop

## 3 Multi-Way Decision (Otherwise If)

### Q7. Grade Classification

**Problem:** Display grade based on marks.

**Algorithm:**

1. Start
2. Take marks as input
3. If marks are less than 0 or greater than 100, display “Invalid marks”
4. Otherwise if marks are 90 or above, display “A+”
5. Otherwise if marks are 80 or above, display “A”
6. Otherwise if marks are 70 or above, display “B+”
7. Otherwise if marks are 60 or above, display “B”
8. Otherwise if marks are 50 or above, display “C”
9. Otherwise if marks are 40 or above, display “D”
10. Otherwise, display “E”
11. Stop

## Q8. Age Group

**Problem:** Categorize a person based on age.

**Algorithm:**

1. Start
2. Take age as input
3. If age is below 13, display “Child”
4. Otherwise if age is between 13 and 19, display “Teenager”
5. Otherwise if age is between 20 and 59, display “Adult”
6. Otherwise, display “Senior”
7. Stop

## Q9. Traffic Signal

**Problem:** Decide action based on traffic light color.

**Algorithm:**

1. Start
2. Take signal color as input
3. If color is red, display “Stop”
4. Otherwise if color is yellow, display “Get Ready”
5. Otherwise if color is green, display “Go”
6. Otherwise, display “Invalid signal”
7. Stop

## 4 Multiple Conditions (Multiple True)

### Q10. Divisibility Test

**Problem:** Check divisibility of a number.

**Algorithm:**

1. Start
2. Take a number as input

3. Check if number is divisible by 3
4. If yes, display “Divisible by 3”
5. Check if number is divisible by 5
6. If yes, display “Divisible by 5”
7. Stop

## **Q11. Number Properties**

**Problem:** Identify properties of a number.

**Algorithm:**

1. Start
2. Take a number as input
3. Check if number is positive
4. If yes, display “Positive”
5. Check if number is even
6. If yes, display “Even”
7. Stop

## **5 Logical Conditions**

### **Q12. Scholarship Decision**

**Problem:** Student gets scholarship if marks are high or sports quota exists.

**Algorithm:**

1. Start
2. Take marks as input
3. Take sports quota status as input
4. Check if marks are at least 85 or sports quota is available
5. If yes, display “Scholarship granted”
6. Otherwise, display “Not eligible”
7. Stop

### **Q13. ATM Withdrawal**

**Problem:** Check whether withdrawal is possible.

**Algorithm:**

1. Start
2. Take account balance as input
3. Take withdrawal amount as input
4. Check if balance is sufficient and amount is a multiple of 100
5. If yes, display “Withdrawal successful”
6. Otherwise, display “Transaction failed”
7. Stop

### **Q14. Online Order Check**

**Problem:** Order is placed only if item is in stock and payment is successful.

**Algorithm:**

1. Start
2. Take stock status as input
3. Take payment status as input
4. Check if item is in stock and payment is successful
5. If yes, display “Order placed”
6. Otherwise, display “Order failed”
7. Stop

## **6 Logical NOT**

### **Q15. Holiday Decision**

**Problem:** Decide whether to go to school.

**Algorithm:**

1. Start
2. Take holiday status as input

3. Check if it is not a holiday
4. If yes, display “Go to school”
5. Otherwise, display “Enjoy your holiday”
6. Stop

## 7 Membership Check

### Q16. Username Validation

**Problem:** Check whether a username exists in the system.

**Algorithm:**

1. Start
2. Store list of valid usernames
3. Take username as input
4. Check if username exists in the list
5. If yes, display “Login successful”
6. Otherwise, display “User not found”
7. Stop